

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 | #include<stdio.h>
2 | int main(){
3 |     int n;
4 |     scanf("%d",&n);
5 |     int count =0;
6 |     count++;
7 |     int c=0;
```

```
7   int c=0;
8   count++;
9   for(int i=n/2;i<n;i++)
10  {
11      count++;
12      count++;
13      for(int j=1;j<n;j=2*j){
14          count++;
15          count++;
16          for(int k=1;k<n;k=k*2){
17              count++;
18              c++;
19              count++;
20          }}}
21   printf("%d",count);
22 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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